

Pawan Kumar

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Synopsis:

Data Migration Specialist with 4+ years of experience in migrating data from Teradata to Snowflake and Azure using advanced ETL tools. Proficient in Snowflake, Python, PySpark, Informatica PowerCenter, SQL, and Control-M for comprehensive data management and transformation. Expert in Advanced SQL and Data Warehousing to enhance reporting and decision-making. Skilled in defect resolution, quality management, and optimizing resources. In-depth knowledge of Snowflake database, schema, and table structures, with expertise in dimensional modeling, including Star Schema and Snowflake Schema.

Experience Summary

- 4+ years of experience in Data Migration from Teradata to Snowflake and Azure.
- Proficient in Advanced SQL, Data Warehousing, and ETL Development.
- Experienced in Azure Databricks, SnowSQL, SnowPipe, and Snowflake tasks.
- Skilled in defect resolution, quality management, and cross-functional team collaboration.

Skills

- Snowflake
- Advance Sql
- Snowsql
- Snowpipes
- SQL server
- Pyspark
- Python
- MySQL
- Data warehousing
- Control-M
- Tidal
- Jira

Qualification

Degree	Major	College	Location	CGPA
Bachelor of Engineering	Computer Science and Engineering (CSE)	Lakshmi Narain College Of Technology	Bhopal, Madhya Pradesh	8.47

Certification

Achievements & Awards
DP-900 Azure Data Fundamentals Certification
DP-203 Data Engineering on Microsoft Azure
SnowPro Associate: Platform Certification
SnowPro Core Certification

Project Experience

Project Name: T-Mobile Telecommunication Data Integration

Role: Data Migration Specialist

Environment: Teradata (TD), Snowflake (SF), Python, Informatica, Control-M, Power BI, SharePoint

Duration: OCT 2020 to NOV 2023

Description:

T-Mobile is a telecommunications project based in the USA focused on migrating data from a legacy system (Teradata) to Snowflake. The project architecture comprises two layers: Stage and Core. Data is received in three formats: File, Database Pull, and Streaming. Informatica, the ETL tool used by Teradata, appends the respective transformations in JSON format before loading them into the Stage table.

Two types of source files, CONTROL and DATA files, are uploaded to SharePoint. Validation is performed on both files using scripts. If validation is successful, the files are forwarded for processing; otherwise, an email notification is sent, requesting a re-submission. The validated data is then loaded into the Staging layer using a one-to-one mapping process that involves truncating and loading the data.

After staging, the data is categorized into Facts and Dimensions and loaded into the Core table. A view is created on top of the Core table, which is integrated with Power BI to provide the client with real-time dashboard views.

Role and Responsibilities:

- Migrated data from Teradata to Snowflake, maintaining a dual-layer architecture (Stage and Core).
- Implemented data ingestion from multiple sources, including File, Database Pull, and Streaming.
- Utilized Informatica to convert ETL transformations into JSON for seamless integration with Snowflake.
- Performed file validation using scripts, automating notifications for failed validations.
- Executed one-to-one mapping to load data into the Staging layer, followed by segregation into Facts and Dimensions for Core layer loading.
- Created views on Core tables, facilitating real-time data visualization through Power BI dashboards.
- Involved in production support by resolving defects such as data mismatches and count discrepancies. Conducted root cause analysis by tracing data issues from Power BI views back to Stage tables.
- Converted Teradata BTEQ scripts into equivalent Python and Shell scripts, maintaining Snowflake compatibility and result accuracy.
- Utilized Control-M as an external scheduler to automate and manage Snowflake jobs, ensuring efficient workflow orchestration.

Project Name: PepsiCo Data Migration and Modern Data Intelligence Platform (MDIP)

Role: Data Engineer

Environment: Teradata, Azure Data Lake, Azure Synapse, Databricks, PySpark, Azure Data Factory (ADF), SQL Server Management Studio, Delta Lake

Duration NOV 2023– Feb 2025

Description:

PepsiCo is an American multinational food, snack, and beverage corporation. This project focuses on offloading analytical and non-analytical workloads from the existing Teradata cluster to an Azure-based Data Lake and Synapse Modern Data Intelligence Platform (MDIP). The objective is to create assets for fetching records from Teradata into Delta tables, referred to as Bronze Merge tables.

Two types of assets are utilized to pull data from Teradata: Full and Incremental.

- **Full Load:** Entire data copy pulled every time. Used when no incremental keys (e.g., last_updt_dttm) are available or when no joins are required on the upsert.
- **Incremental Load:** Data is loaded incrementally using watermark columns. Preferred for large datasets with high data frequency.

The project involves multiple layers:

- **Acquisition (ACQ) Layer:** Data is built through files or external sources.
- **DWL Layer:** Data flows from ACQ to DWL through BTEQ scripts, which are converted to PySpark using the CDIT team's framework.

- **S&T Framework:** A PepsiCo-specific UI-based framework that manages read access and stores data in Azure Gen2 Storage in Parquet format.

Data flows from ACQ to Stage, and then from Stage to Load, enabling seamless integration and processing within the Azure ecosystem.

Role and Responsibilities:

- Analyzed existing data structures, schemas, and dependencies in Teradata to plan and execute migration to Azure.
- Developed Databricks notebooks using PySpark transformations and DataFrames for efficient data processing.
- Designed scripts using Spark SQL for enhanced performance in Azure Databricks, converting BTEQ (Basic Teradata Queries) to Spark-based transformations.
- Processed data in the DWL layer, storing it in Azure Data Lake Storage (ADLS) for downstream applications.
- Collaborated with Azure Data Factory (ADF) to trigger and schedule CI/CD pipelines for automated notebook execution.
- Monitored, designed, and triggered Data Pipeline jobs in Azure Data Factory for efficient workflow orchestration.
- Performed Unit Testing with QA and Production data to ensure successful migration from Teradata to ADLS.
- Implemented Data Lake Architecture using Delta Lake, enabling efficient storage management and querying of structured and semi-structured data.
- Utilized PepsiCo's proprietary framework, S&T, to move raw data to the Acquisition layer, ensuring data integrity and accessibility.

Work Experience

Cognizant Technology Solutions

Role: Associate Developer

Duration: Oct 2020 – Feb 2025

Worked as an Associate Developer, contributing to data migration and integration projects with a focus on Teradata to Snowflake transformations. Collaborated with cross-functional teams to design and implement data pipelines, optimize ETL processes, and ensure data accuracy and integrity. Involved in defect resolution, quality management, and production support, leveraging advanced SQL, Python, and Control-M for automation and efficiency.

Personal details:

Name: Pawan Kumar

Do you have a Passport: Yes

Visa Status: N/A

Location: Pune